



Low-Cost Marine Recorder (LC-MARE) Building your own underwater sound recorder

Objective

- Foster innovation and broadening participation in marine research conservation, and restoration
- Provide a cost-effective, open-source solution for accessing and recording underwater sound.
- Implement a low-cost solution without limitations on sound quality (science-grade data).

Concept

- Use commercially available components were possible
 - PVC pressure tubes for underwater housing
 - Digital recording components
 - Hydrophones
- Use custom components were necessary
 - Analog front-end (preamp and ADC)
 - Hydrophone mounting
- Avoid use of expensive machinery
- Calibration to allow comparisons and exchanges of data
- Restrict use cases to coastal activities or shallow deployments
- Minimize power consumption to increase endurance
- Open-Source Hard and Software (QR code top-right)

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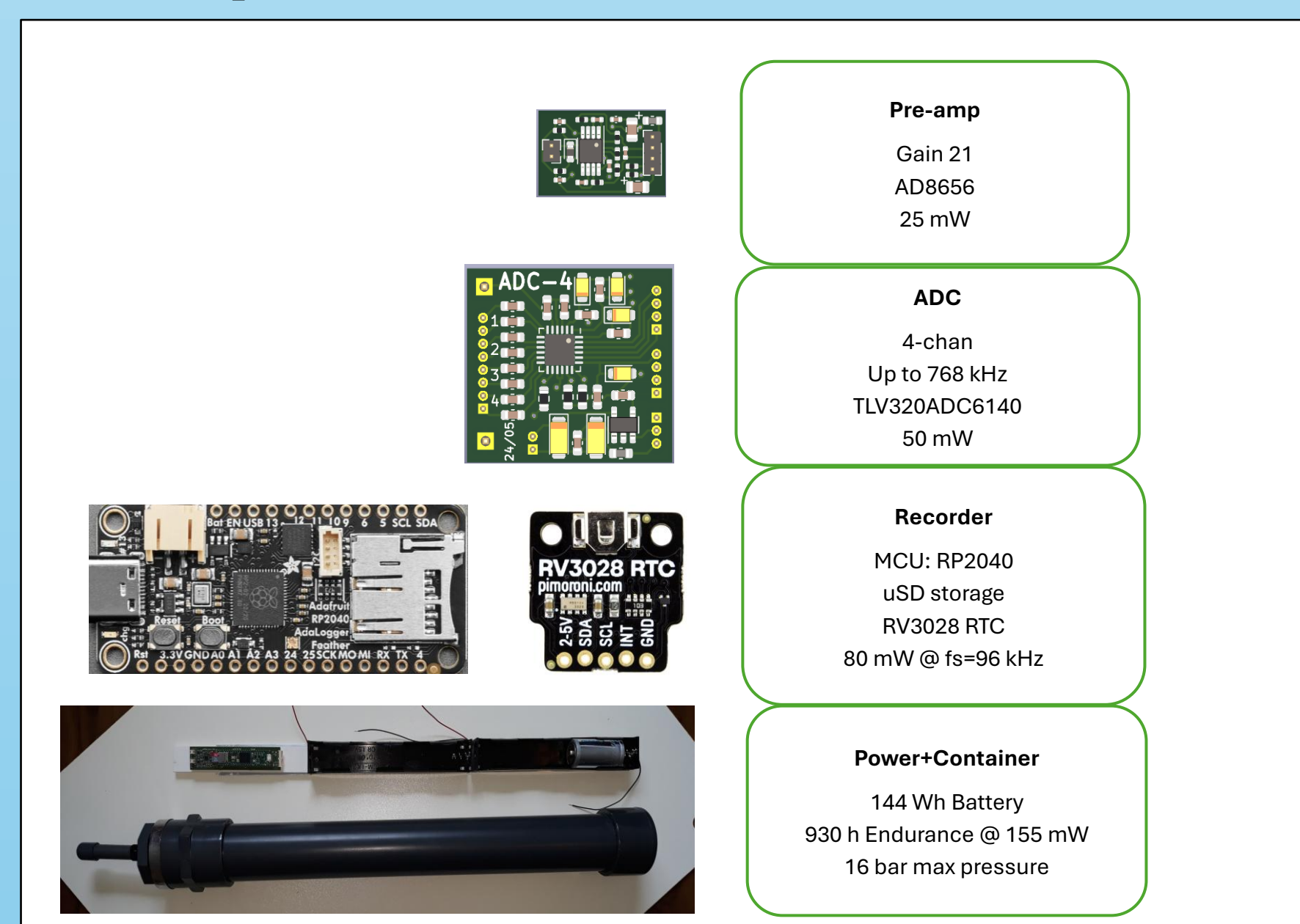
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Use cases

- Generation of science-grade marine acoustic data for supporting entry-level bioacoustics scientific research (e.g. for undergraduate and graduate students)
- Environmental monitoring of coastal marine areas (being protected or not) up to a sensor depth of 100 m
- Acoustic recorders for underwater vehicles
- Promotion of marine bioacoustics in STEM education

Components



Hydrophones

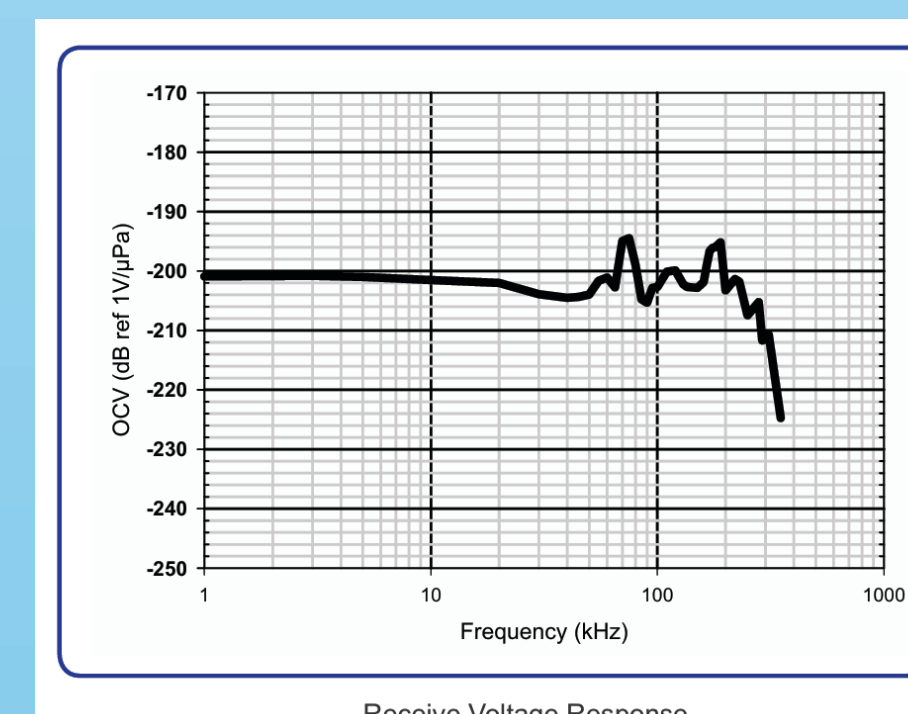


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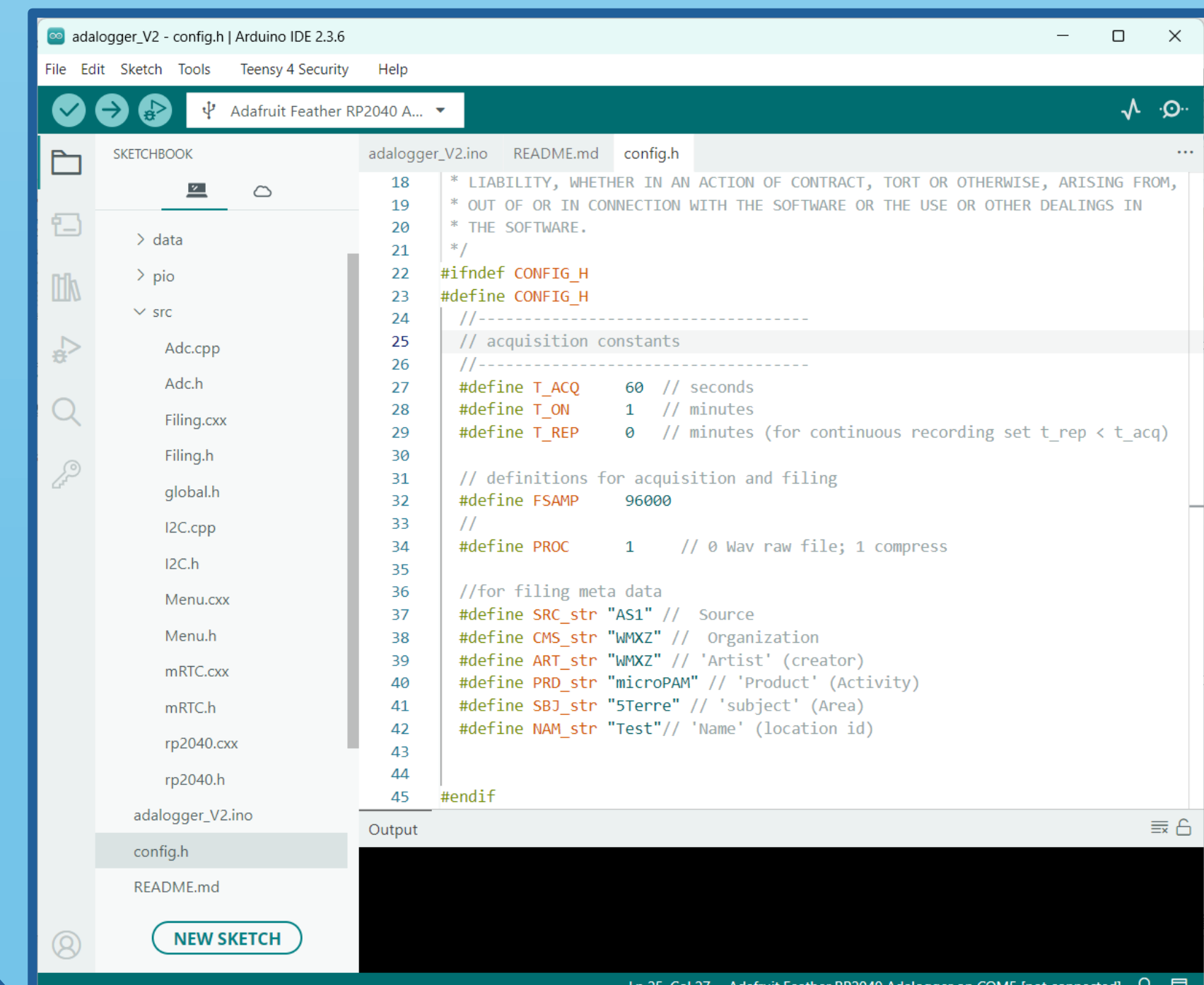
Sensortech (CA)

S = -205 dB//1V/Pa

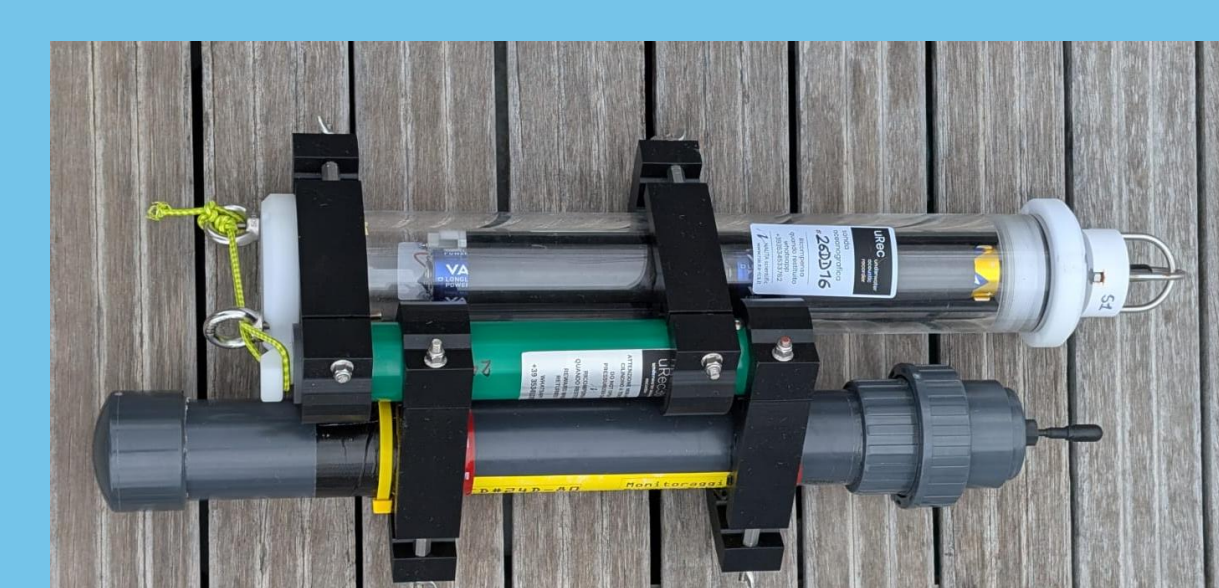
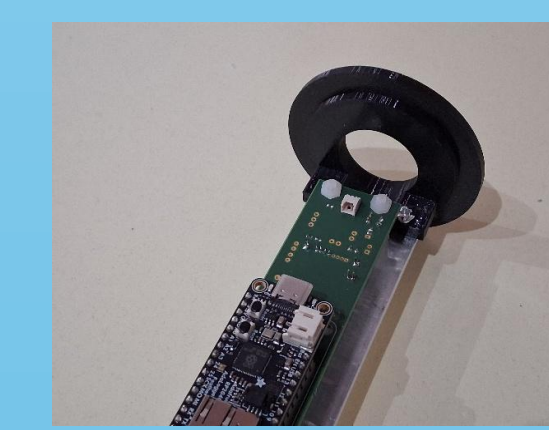
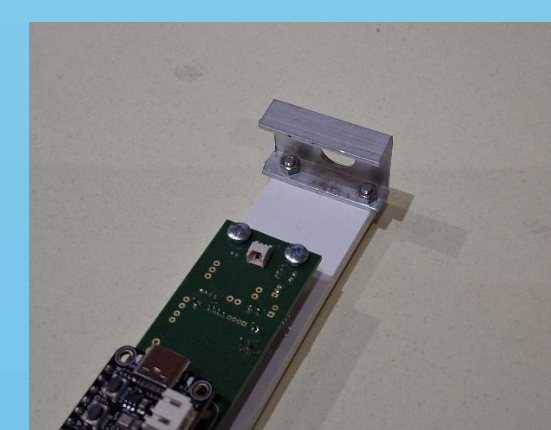
C = 1.6 nF



Software



Construction options



Comments

- Building a science-grade high-performing underwater sound recorder is possible without the use of expensive machining tools.
- Essential for the success is the limitation of use cases to recorder depth of up to about 100 m, suitable for PVC pressure tubes.
- Using a rectangular form factor of the electronics and sequential arrangement of batteries are necessary to limit the diameter of the pressure containers.
- The use of COTS electronic components and modular approaches helps to experiment with custom modifications of the components.
- The use of custom components for pre-amplifiers and Analog-Digital-Converters is affordable if obtained in larger quantities suggesting coordinated efforts.
- Different mounting options of the electronic pack allow flexible adaption to the availability of pressure housing components.
- Use of 3D printer eases mounting of electronics and battery packs
- Software development using the Arduino infrastructure allows easy adaptations of firmware to use-case specific requirements.
- Hydrophones are still the most expensive part of recorder but coordinated procurement efforts can reduce costs.
- Multiple options to assemble hydrophones allow flexible adaption to the availability of components (e.g. pressure container penetrator)
- The actual material cost of LC-MARE is about 320.0 € (Electronics: 110 €; Pressure container: 20 €; Hydrophone SQ17: 190 €)